Non-GUI codes

To understanding the code and the model first look at the non-GUI codes. These consist of the following scripts:

boulet_model_main (bulk of the code)
boulet_stimulation (called by main)
boulet_parameters (called by stimulation)
random_correlated (small script to create correlated random values, needed in parameters)

Called after boulet_model_main generates its results (spiketimes):
ECAP_plot_manual (generates ECAP response by summing ECAP responses to single stimulus pulses)
ECAP_amplitude2 (calculates the ECAP for a single stimulus pulse)
UR.m (defines the unit action potential shape, used to calculate the ECAP)

After understanding these codes, take a look at *script_recoveryfun* and other *script_(something)* scripts, which are used to automate the process of obtaining the recovery function and faciliation/accommodation function. And if you are curious, a computational Hodgkin-Huxley is also available (*HH_model*) for comparison of „real“ and modelled neural behaviors (excluding SRA); this model was used to generate the figures extensively used in the introductory part of the report.

The *boulet_model_main* script has three further variants called *boulet_model_light (1, 2 or lighter)*, which increase computational speed by not saving unnecessary parts of the data (the only crucial data is the timings of spikes), as discussed in the report. Some of these modifications also affect the parameters (noise variable) and the stimulus variable, hence *boulet_parameters_light* and *boulet_stimulation_light* scripts are used some of the „light“ versions of the model.

GUI codes

GUI codes are identical to the non-GUI codes, but with few crucial differences. First, any script requiring an input (e.g. model requires the stimulus and parameters) is written as a function; and second, all parameters and data are saved as structures (e.g. handles.parameters.nneurons), which simplifies reading the code, finding manually the value of some parameter, and adding new model features (since these simply get added into the structure and no futher modifications of code are needed).

Similarly, since values of parameters can be changed in GUI, the model itself cannot be called directly, but an IFmain function is called where input-dependent variables (e.g. time vectors, which depends on timestep) are first calculated, and only then is the model executed and ECAP values calculated. Therefore, the code structure is:

mainGUI -> IFmain -> IFmodel -> IFstimulation

with the exception that initial values of parameters in GUI and the corresponding ECAP require calling the functions *initIFparameters* and *initIFstimulation* (later on these functions are not used).

As before, light versions of the IFmodel exist, and calculation of the AGF and recovery/accommodation function has been automated by using *AGF_ECAP* and *recovery_function_ECAP*. If necessary, one can check which neurons have fired in response to each pulse by using the whofired script – this script currently operates „offline“ (not in GUI) and should be included in GUI and its plot shown as it gives valuable information.

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